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Complex indefinites and the projection of DP in Mandarin Chinese

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Abstract This paper argues for the projection of DP for indefinites initiated with *yi* 'one' and indefinites initiated with a classifier in Mandarin Chinese. First, a null numeral for 'one' is identified. Second, in *yi*-initial nominals, an indefinite article is also identified. Third, for classifier-initial indefinites, which have no overt numeral, a head movement of a classifier to D is argued for. Therefore, the head of DP in the language can be realized by either an indefinite article or a raised classifier. No argument is in the structure of a bare Classifier Phrase in the language. The paper supports the idea that numeralless classifier nominals exist because of the availability of null numerals. It also proposes a semantic LF reanalysis approach to the dependency of a classifier-initial nominal on a higher head element in the language.

Keywords Indefinite · Classifier · Article · DP · Proform · Numeral · Mandarin

1 Introduction

Syntactic arguments in Mandarin Chinese (MC), a numeral classifier (CL) language, can be in either a simple form (i.e., a bare noun) or a complex form. This paper discusses the syntax of indefinites in complex forms in the language. Cheng and Sybesma (1999: 530) claim that indefinites in MC are NumeralPs instead of DPs, and Bošković (2008, 2012) argues that article-less languages have no DP. On the other hand, Jenks (2018) shows that for definites, bare nouns are used for unique definites and complex forms are used for anaphoric definites in MC, as well as in

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some other languages. A similar contrast in other CL-languages is reported in Simpson and Biswas (2016) and Simpson (2017). Jenks further argues that bare noun forms are just NPs and complex forms are DPs. Thus, the NP-DP division can be cut with respect to different structures in the same language and the same conditions on the division can be found in both CL- and non-CL-languages. Importantly, according to Jenks's study, DP is projected for definites in MC. With this background, this paper explores complex indefinites in MC, to answer the question of what the syntactic categories of complex indefinites are in the language.

As generally assumed, a bare noun nominal is one that does not contain any of three elements: an overt D-domain element, such as an article or demonstrative, a CL, and a numeral. Thus, the post-verbal nominal in (1a) is a bare noun nominal, but those in (1b) and (1c) are complex ones.

(1) a. Alin xiang mai shu.[1]
Alin want buy book
'Alin wants to buy {the book(s)/books/a book}.'
b. Alin xiang mai ben shu.
Alin want buy CL book
'Alin wants to buy a book.'
c. Alin xiang mai yi ben shu.
Alin want buy one CL book
'Alin wants to buy a book.'

Complex indefinites and bare noun indefinites do not have the same distribution. On the one hand, in many languages, in contrast to a bare noun indefinite, a complex indefinite cannot be a kind-level argument. The complex indefinite subject in (2b) is not acceptable, compared to the bare plural subject in (2a) (Lawler 1973: 110; see Krifka 2004, Dayal 1992, 2004, Chierchia 1998, for accounts). Similarly, in MC, the kind-level argument is a bare noun in (3a) (also see Krifka 1995: 398 (1a)), but the complex form in (3b) is not acceptable.

(2) a. Dogs are common.
b. *A dog is common.

(3) a. Shi-hu kuai juezhong-le.
stone-tiger soon extinct-PFV
'Leopard cats will soon go extinct.'
b. *Yi zhi shi-hu kuai juezhong-le.
one CL stone-tiger soon extinct-PFV

On the other hand, in many languages, in contrast to a complex indefinite, a bare noun indefinite cannot express a specific reading. For instance, (4a) does not mean that certain spiders were not found by Yella. Thus, in English, bare plurals do not

[1] Abbreviations: CL: classifier; BA: causative; DE: modification/nominalization; EXP: experiential; IMP: imperative; PFV: perfective; PRG: progressive; PRT: sentence-final particle; Q: question.

have a specific reading (Carlson 1977; Diesing 1992: 64; Wilkinson 1996: 213; Kratzer 1998: 171).

(4) a. Yella didn't find spiders in her bed. b. Everyone read books about slugs.

The same is true in MC (e.g., Lee 1986: 193, Huang 1987, Liu 1997), and thus the positions that reject nonspecific nominals reject bare noun indefinites. One such position is to the immediate right of the functional element *ba*. This position allows a complex indefinite (e.g., Lü 1944 [1990: 161]), and the indefinite has a specific reading only (Chen 2004: 1170; Zhang 2013: 145; Jiang 2015a: 335), as seen in (5a). In this example, if the complex indefinite *(yi) ge hao pengyou* 'a good friend' is replaced with the bare noun nominal *hao pengyou* 'good friend', the latter has a definite reading only.[2] Another position is in a construction where an indefinite is followed by an apparent attributive. The indefinite must be in a complex form and it has a specific reading exclusively (Huang 1987), as seen in (5b) (it can be a CL-initial one; see Zhang 2008: 18; 2013: 146; Jiang 2015a: 336).[3] Thus, neither the distribution of definite bare nouns (Jenks 2018) nor that of indefinite bare nouns is free.[4]

(5) a. Ta ba (yi) ge hao pengyou gei dezui le.
he BA one CL good friend give offend PFV
'He got a good friend offended.' (see Huang 2015: 35)

b. Ta jiao-le {*nü-pengyou/(yi) ge nü-pengyou} hen
he make-PFV girl-friend/one CL girl-friend very
piaoliang.
pretty
'He has a girlfriend who is very pretty.' (Jiang 2015a: 336)

The contrasts between bare noun and complex indefinites justify a thorough study of each type. In this paper, I discuss the syntax of the complex type only, including CL-initial indefinites, as in (1b), and *yi*-initial ones, as in (1c).

In MC, *yi* 'one' can encode the numeral for 'one', in contrast to other numerals. All numerals in numeral-initial nominals may have an "exactly" reading in certain contexts (called a quantity reading in Li 1998; see Tsai, to appear). *Yi* can also encode an indefinite meaning, in contrast to a definite reading (Chen 2003: 1171). I ignore other readings of *yi* here (see Chen 2003, Zhang 2013: 93, among others).

[2] In Hsueh (1987) and Cui (1995: 17), the post-BA nominal is a topic and the VP to its right is the comment, and the VP is focused. The referent of the nominal is known to the speaker. Zhang et al.'s (2017) experimental phonetic study supports this analysis. Since a topic must not be nonspecific in MC, the post-BA nominal rejects a nonspecific reading.

[3] In Zhang (2008), in the construction represented by (5b), the post-verbal indefinite is a topic and the expression to its right (called coda, e.g., *hen piaoliang* 'very pretty' in (5b)) is the comment. We find that the combination of the two elements typically expresses a categorical judgment. Since a topic must not be nonspecific in MC, the indefinite nominal in the construction rejects a nonspecific reading.

[4] A common misunderstanding is seen in Delfitto & Fiorin (2017: 2): "[t]here are languages, such as Chinese, where the occurrence of bare nouns (BNs) in argument position is virtually unconstrained".

Focusing on the two basic types of complex indefinites, I do not discuss nominals that have a numeral other than *yi* 'one'. I also do not discuss other complex nominals such as possessive nominals.[5]

I show that *yi*-initial indefinites can be DPs in which *yi* is an article (Sect. 2), and CL-initial indefinites are DPs in which the CL has moved from CLP to D (Sect. 3). Thus, in addition to being the functional category that hosts a numeral (but not bare CLP), DPs can be categories of complex indefinites in MC.

2 *Yi*-initial nominals

Four issues are discussed in this section: where would certain functional words be hosted if there were no DP in MC (Sect. 2.1), what happens if certain kinds of nominals have a consistent singular reading without a numeral for 'one' (Sect. 2.2), where are the syntactic positions of numerals and CLs in MC (Sect. 2.3), and can *yi* 'one' be an indefinite article (Sect. 2.4)?

2.1 *Mou* 'certain' and *na* 'which'

In MC, an indefinite may start with the word *mou* 'certain' or the interrogative *na* 'which'.

(6) a. mou yi ge zi
certain one CL character
'a certain character'

b. mou ji ge zi
certain several CL character
'certain characters'

(7) a. na yi ge zi
which one CL character
'which character'

b. na ji ge zi
which several CL character
'which characters'

Mou is a specific indefinite marker (Liu 1997), like *a certain* in English (Fodor and Sag 1982, Hintikka 1986). It is not an adjective, since it never occurs with the particle *de*, whereas adjectives allow or require *de*. *Na* 'which' is also an indefinite marker, although it is called a demonstrative determinative in Chao (1968: 565). Also, *mou* and *na* are not specified with any number or quantity information. Thus, in (6a), since the numeral *yi* 'one' occurs, the nominal is singular, and in (6b), where the vague plural numeral *ji* 'several' occurs, the nominal is plural.

In English, the string *a certain* is analyzed as a complex determiner in Kratzer (1998: 168), and the word *which* is analyzed as the WH counterpart of a determiner or demonstrative in the literature (e.g., Adger 2003: 346; also see Liu 1997: 153 for discussion and references). Where are the syntactic positions of *na* and *mou*, which signal the indefinite status of the hosting nominal in MC? Neither NP nor Cheng and

[5] The CLs discussed in this paper are not restricted to *ge*. The CL *ge* has some special uses, e.g., it may occur in a secondary predicate (see Lü 1990 [1944]: 153–154, Zhang 2016). This paper does not discuss any of these special uses.

Sybesma's (1999: 528) NumeralP for indefinites in MC is able to accommodate these elements. They must be hosted by DP. I assume that they are merged as SpecDP (Liu 1997: 68–69 calls *mou* a logical determiner). Thus, at least for *mou*- and *na*-initial indefinites, DP is attested in this CL-language.[6] However, other forms of complex indefinites can also be DPs, as argued in the remaining part of the paper.

2.2 The null proform of the numeral for 'one'

I argue that in addition to the overt numeral *yi* 'one', MC also allows a silent proform for this numeral. A CL may directly follow a demonstrative (Chao 1968: 565–566), as in (8a), although it usually follows a numeral, as in (8b). The words *mei* 'every', *mou* 'certain', and *na* 'which' are also followed by an optional *yi* in (9).

(8)	a.	zhe	ben	shu	b.	zheyi	ben	shu
		DEM	CL	book		DEMone	CL	book
		'this book'				'this book'		

(9)	a.	mei	(yi)	ben	shu	b.	mou	(yi)	ben	shu
		every	one	CL	book		certain	one	CL	book
		'every book'					'a certain book'			
	c.	na	(yi)	ben	shu					
		which	one	CL	book					
		'which book'								

However, examples like (8a) and the *yi*-less versions of the examples in (9) reveal a semantic constraint: they must denote a singular individual or singular unit specified by the CL. This is pointed out by Greenberg ([1972] 1990: 188): "in Mandarin the classifier *ben* required with *shu* 'book' with any number (e.g., *i ben shu* 'one book', *san ben shu* 'three books') occurs with the demonstrative also (*che ben shu* 'this book') but only in the singular." Unlike *this* in English, *zhe* 'this, these' does not encode a singular meaning, and thus it can be followed by any numeral, in addition to *yi* 'one' (e.g., *zhe san ben shu* 'these three books'). Similarly, *mei* 'each' and *na* 'which' can also occur with any numeral. In the absence of a numeral, such complex nominals consistently express a singular meaning. Therefore, there must be a singular-denoting element in these forms.

The element appears to be either a deleted *yi* (cf. Huang 2015: 38 fn. 37), or just a null proform for 'one' (Cheng and Sybesma 1999: 530; cf. Huang 2015: 38 fn. 37). Both hypotheses are able to explain the following facts. First, the numeral *yi* occurs

[6] In (ia), *mou* is combined with a noun, but such a combination is not productive, as seen in (ib) and (ic). Unlike a bare noun, (ia) must be indefinite and singular. Thus, even for (ia), the indefinite reading must be hosted in a DP.

(i)	a.	mou	ren	b.	*mou	shu	c.	*mou	pingguo
		certain	person		certain	book		certain	apple
		'a person'							

in the syntactic structure of such nominals, and thus their singular meaning is captured. Second, the deletion of *yi* or the null *yi* seems to be optional, since an overt *yi* is available in the syntactic position, as seen in (8b) and (9). Third, if an element is focused or foregrounded, it cannot be deleted or surface as a null form. The same is true of the numeral for 'one'. In (8a), the singular meaning is not foregrounded or focused. The deleted or null *yi* is never associated with a focus marker, as seen in (10b), in comparison with (10a), which has the overt numeral *yi* 'one'. In (10b), *zhi* 'only' can be associated with the demonstrative *zhe* 'this', and thus the first clause contrasts with clause α, which contains *na* 'that'. This focus marker can also be associated with the noun *shu* 'book', and thus the first clause contrasts with clause β, which contains *zazhi* 'magazine'. However, the focus marker is not associated with the implicit numeral for 'one', and thus the first clause may not be followed by clause γ, which contains the contrastive *san* 'three'. Since no deleted or null element is focused, the silent *yi* cannot be focused, as expected.[7]

(10)	a.	Alin	zhi	mai-le	zhe	yi	ben	shu,	bushi	san	ben.
		Alin	only	buy-PFV	this	one	CL	book	not	three	CL

'Alin bought only this one book, not three.'

b.	Alin	zhi	mai-le	zhe	ben	shu,	{α bushi	na	ben
	Alin	only	buy-PFV	this	CL	book	not	that	CL
	shu/	β bushi	zhe	ben	zazhi/	γ #bushi	san	ben	shu.
	book/	not	this	CL	magazine/	not	three	CL	book

'Alin bought only this book, {α not that one/β not this magazine/γ #not three}.'

Nevertheless, there are reasons to reject the *yi*-deletion hypothesis and posit a null *yi*, which I label YI_{pro}. First, ellipsis needs an antecedent (Chomsky 1965: 144). (11) shows that if *san* 'three' occurs in the first clause, qualified to be an antecedent for an assumed deletion, the missing numeral in the second clause should be *san*. However, the implied numeral meaning in the second clause is consistently 'one' (cf. Zhang 2013: 140).

(11)	Alin	mai-le	zhe	san	ben	shu,	Ajiao	mai-le	na	ben	shu.
	Alin	buy-PFV	this	three	CL	book	Ajiao	buy-PFV	that	CL	book

'Alin bought these three books, and Ajiao bought that book.'

[7] A demonstrative may also combine with a reduced *yi* 'one', deriving *zhei* (*zhe* + *yi* 'this + one') and *nei* (*na* + *yi* 'that + one') (e.g., Cheng and Sybesma 1999: 530 fn. 17). However, because of reanalysis, *zhei* and *nei* can also function as pure demonstratives, followed by a numeral other than *yi* 'one'. So (i) is acceptable (Bisang 1999: 145). The opaque function of *yi* in *zhei* in (i) is similar to the opaque function of *et* 'and' in the phrase *and etcetera*, where two conjunctions (*and, et*) occur in a row (the Latin word *cetera* means 'the rest') (see Zhang 2013: 144).

(i)	*zhei*	*san*	*feng*	*xin*
	DEM	three	CL	letter

'these three letters'

Similarly, as noticed by Kayne (2012: 78), (12b) is acceptable, but not under the reading of (12a). Thus numerals cannot be deleted. Similar examples are seen in Italian (Cinque 2012: 180) and Naxi (Law 2012: 112–115).

(12) a. Maryhas written four papers, whereas John has only written four squibs.
b. Mary has written four papers, whereas John has only written squibs.

On the other hand, like the *pro* argument in the language, YI_{pro} is also discourse-oriented (i.e., it is a radical *pro*-drop). Unlike an elided element, it does not need a linguistic antecedent. Instead, it is licensed whenever the meaning of 'one' is obvious to the speaker.

Second, deletion needs a structural licensor, which is usually a head element next to the elided part. Even if we assume that the elements to the left of *yi* in (8) and (9) are head elements, we never see them license the ellipsis of another kind of element. Those pre-*yi* elements are not ellipsis-licensors. On the other hand, some head elements may license ellipsis, but they never license the deletion of *yi*. In (13a), *de* licenses the deletion of *xuesheng* 'student', but *de* cannot license the deletion of *yi*, as seen in (13b) (adapted from Li and Bisang 2012: 346).[8]

(13) a. Wo zai zhao [xue yingyu de xuesheng].
I PRG seek study English DE student
'I am looking for students who study English.'

b. Wo zai zhao [xue yingyu de *(yi) ge xuesheng].
I PRG seek study English DE one CL student
'I am looking for a student who studies English.'

Third, the *yi*-deletion hypothesis stipulates that only *yi*, but not any other numerals, can be deleted (Yang 2001: 71); however, *yi* may not be deleted in other contexts, such as fractions (*san-fen-zhi-*(yi)* 'one third') and ordinals (e.g., *di *(yi) bei shui* 'the first cup of water'), as pointed out by Zhang (2013: 141). In contrast, cross-linguistically, the numeral for 'one' may have an obligatory null form in complex numerals (Hurford 1987; Watanabe 2014), and it may have a weak form, in addition to a strong form, in the same language. In Malay, the numeral 'one' has both a free and a bound form: *satu* and *se*, respectively (Nomoto 2013: 13). The numeral for 'one' also exhibits special morphological properties different from other numerals in languages such as French and Russian (Hurford 2002: sec. 5). YI_{pro}, as a weak form of the numeral *yi* in MC, is thus not ad hoc.

Fourth, not every CL language allows a CL to occur without a numeral. Cantonese has parallels of (8) and (9) (Sze-wing Tang, p.c., Huang 2015). Zhuang (Ma 2003: 711), and Thai, Lao, Bouyei, Maonan (Lu 2012: 146–148), Nuosu Yi

[8] Jiang (2012: 201) shows that the obligatory occurrence of *yi* 'one' in (13b) is not related to the specificity of the complex nominal (contra Li and Bisang 2012). She suggests that *yi* cannot be deleted here, because *ge xuesheng* cannot be reconstructed back into the subject position of the relative clause (**ge xuesheng xue yingyu*). However, what is reconstructed should be *xuesheng* alone; also, it is not clear why PF deletion needs to consider reconstruction. See Sect. 3.4 for an account of (13b), and see Jiang (2012, 2018) for more discussion of *yi*-deletion.

(Jiang 2018: 6), Bangla (Dayal 2012: 214), and Vietnamese (Nguyen 2004: 61) also allow the direct combination of a demonstrative with a CL, without a numeral. However, Korean, Japanese (Yi 2011: 268), and Min (Chen 1958: 503; Huang 2015: 38 fn. 37) reject such forms. For the latter type of language, the information-structure-driven deletion assumed by the deletion hypothesis does not exist. From the perspective of YI_{pro}, such languages do not have a null pro-form for the numeral for 'one', making this similar to other cross-linguistic contrasts, which usually involve contrasts in lexical specifications (cf. Borer 1984), rather than from the availability of a deletion operation under the same information structural considerations.[9]

My YI_{pro} approach also differs from that of Akhenvald (2000: 183), where it is suggested that the CLs with demonstratives and without any overt numeral in MC can be "deictic" CLs (I thank Byeong-uk Yi for the reference). This would mean that every CL would have a deictic version in MC. In my approach, it is the form of the numeral, rather than the CL, that is special in the constructions. Akhenvald's claim does not explain why deictic CL nominals are exclusively singular. The deictic sense and the singular meaning are both captured by my YI_{pro}, which takes the 'one' meaning in the discourse context as its antecedent. My approach also differs from that of Yang (2001: 72), who claims that when a CL follows an overt numeral, it is a suffix, but when it follows another kind of element, it is an enclitic. However, if a form is either an affix or clitic, the two morphological uses correlate with different semantics and thus different base positions in syntactic structure. For example, the plural suffix—*s* and the possessive enclitic—'*s* in English have different syntactic positions. There is no evidence to show that the two uses of a CL correlate with different base positions of the CL in syntax.

The above argumentation leads to the recognition of YI_{pro}. The syntactic position of YI_{pro} is the same as that of its overt counterpart.

Cross-linguistically and even within the same language, numerals can be either in a head position or in a Spec position (e.g., Danon 2012; Simpson and Syed 2016; Ionin and Matushansky 2019). In MC, first, unlike some head elements such as verbs, and like phrasal elements such as adverbs, a numeral never licenses ellipsis, as shown in (14a) (Zhang 2013: 215, 253). Second, like a phrasal element, a numeral can be modified, not only in English (e.g., *exactly sixty books*), but also in MC, as seen in (14b) (Zhang 2013: 214). Third, like a nominal phrase, a numeral can be the complement of a preposition, as seen in (14c). Fourth, like a phrasal element, a numeral can be a subject or predicate, as seen in (14d) and (14e), respectively.

[9] In addition to the numeral for 'one', a CL can also be null in a CL language (Simpson and Ngo 2018: 228).

(14) a. Ta mai-le liu ben shu, wo ye mai-le liu *(ben shu).
he buy-PFV six CL book I also buy-PFV six CL book
'He bought six books, and so did I.'

b. jiangjin liushi ben shu
almost sixty CL book
'almost sixty books'

c. cong yi dao wu
from one to five
'from one to five'

d. Yi bi ling da.
one than zero big
'One is bigger than zero.'

e. Jisuan de jieguo shi shiyi.
count DE result be eleven
'The result of the counting is eleven.'

I thus claim that a numeral is phrasal in MC, and is base-generated at a Spec position, as assumed by Li (1999: 87) and Borer (2005: 96), among others. It is a nominal phrase in MC (cf. Ionin and Matushansky 2019), and $\mathrm{YI}_{\mathrm{pro}}$ is a subtype of *pro*, which "is a proform that minimally consists in the categorizing head n, lacking a root" (Barbosa 2019: 1).

2.3 The syntactic positions of numerals and CLs

In MC, a numeral is usually followed by a CL or another kind of unit word (I only discuss CLs in this paper). Following Krifka (1995), and Zhang (2013: 213–216), among others, discusses the semantic and syntactic dependency of a numeral on a CL in the language (also see Bale and Coon 2014), and claims that the dependency should be represented by a Spec-Head relation. However, when a phrase and a head element are semantically and syntactically related, the former does not have to be the Spec of the latter. For example, an agent-oriented adverbial is semantically and syntactically dependent on an action-denoting verb, but the former is not the Spec of the latter, although it c-commands the latter. Such an adverb can be base-generated higher than the Spec of the VP headed by the verb (cf. Travis 1988; Li 2006: 410). There is indeed evidence showing that the syntactic relation between a numeral and a CL is not a Spec-head relation in MC: they can be separated by another element, such as *da* 'big' and *xiao* 'small'. In (15), *da* intervenes between *san* 'three' and the CL *tiao*. Since the Spec-head relation may not be interrupted by any element, a numeral and a CL do not seem to have this relation.

(15) san da tiao yu
three big CL fish
'three big fish'

The element between a numeral and a CL, such as *da* in (15), is a degree element, and it scopes over the unit word and the NP exclusively (Luo et al. 2017). I have found that such a degree element never scopes over a numeral. The meaning of (15) must be that each fish, rather than the combination of three fish, is big. If *da* scoped over the numeral, the following reading would be allowed: one or two of the three fish are small, but the combination of the three is big. The fact that this reading is out indicates that the numeral must be base-generated higher than the degree element.

Unlike adjectives, pre-CL degree elements form a closed class. They select a gradable CL and project the functional DegP (Luo et al. 2017). Thus, they are not a modifier of the CL, contra Zhang (2013: 230) and Jin (2013: 201). Zhang claims that the pre-CL element undergoes a morphological merger with the CL, and Jin, following Tang (1990), claims that it forms a compound with the CL. These modification theories are not compatible with the DegP analysis.

As a functional head element, the pre-CL degree element is morphologically light: it must be monosyllabic (see Tang 1990; Zhang 2013: 229). I have also observed that it is a bound form, taking a monosyllabic word to its left as its phonological host, and therefore it behaves like an enclitic type of functional element. In (16a), *xiao* 'small' cliticizes onto *yi* 'one'. But in (16b), *xiao* is left-adjacent to the disyllabic *shiliu* 'sixteen', and the example is not acceptable.

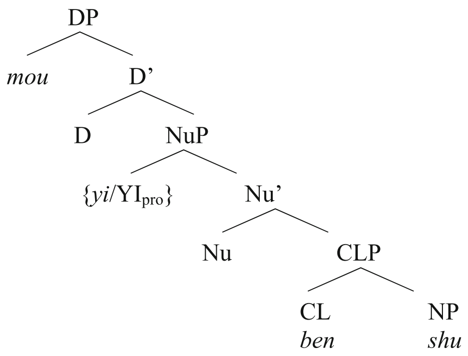
(16) a. na yi xiao zhang zhaopian
that one small CL photo
'that small photo'
b. *na shiliu xiao zhang zhaopian
that sixteen small CL photo

Some types of clitics are very selective of the formal properties of their phonological host, which need not also be their syntactic host (Klavans 1985). This is the case for degree elements for CLs in MC, which require a monosyllabic word to be their phonological host. The enclitic nature of the degree elements shows that the elements are unlikely to form compounds with CLs (contra Jin 2013), since no part of a compound takes a compound-external element as its phonological host.

Semantically, numerals introduce a cardinality function, independent of the functions or properties of other elements of a nominal (e.g., Munn and Schmitt 2005: 825, cited in Espinal 2010: 987; Zamparelli 2005: 924). Adopting Cheng and Sybesma (1999: 529), Li (1999: 87), Huang (2015: 37), and Simpson and Syed (2016: 758), among others, I thus assume that a numeral is hosted in a projection, NuP (Nu = Numeral), higher than the one that hosts a CL. Specifically, I claim that a numeral is at SpecNuP and a CL heads CLP in MC. DegP, if it occurs, is projected between NuP and CLP. To identify a syntactic projection, evidence for either a Spec or head element is sufficient. As for the relation between a CL and its associated noun, since the former selects the latter (e.g., Zhang 2013: 163; also see Simpson and Ngo 2018), they have a head-complement relation. Thus, (15) has the structure in (17). Accordingly, the position of YI_{pro} is SpecNuP. The structure of (9b) is (18), where no DegP is projected.

(17) $[_{\text{NuP}}$ *san* $[_{\text{Nu'}}$ ∅ $[_{\text{DegP}}$ $[_{\text{Deg'}}$ *da* $[_{\text{CLP}}$ $[_{\text{CL'}}$ *tiao* $[_{\text{NP}}$ *yu*$]]]]]]]$ (for (15))

(18) (for (9b))



2.4 The unstressed initial *yi* as an indefinite article

In this section, I argue that the overt *yi* 'one' in MC can be used as an indefinite article. Lü (1990: 174; adopted in Chen 2003: 1171) claims that the cluster of *yi* plus a CL is an article. However, it is unlikely that every *yi*-CL cluster is an article, since there are a lot of CLs in the language. More plausibly, if *yi* occurs at the left edge of a nominal, it alone can be an indefinite article. I label an article use of *yi* YI$_{\text{D}}$. YI$_{\text{D}}$ can be identified by five diagnostics.

First, generic arguments of distributive predicates reject a numeral but allow an article (see Krifka et al. 1995: 35 for possible plural numerals in generic arguments of non-distributive predicates, e.g., *Two magnets either attract or repel each other*). In (19a) and (19b), *a* cannot be replaced by the numeral *one*. In the similar MC examples (19c) and (19d), *yi* cannot be a numeral either. In the generic arguments, neither *a* in (19a, b) nor *yi* in (19c, d) contrasts with a real numeral. *Yi* in this use plays the same role as an indefinite article.

(19) a. A lion is a ferocious beast. (Huddleston and Pullum 2002: 372)

b. A plane travels faster than a train.

c. Yi jia feiji de sudu yao bi yi lie huoche de
one CL plane DE speed will than one CL train DE
sudu kuai.
speed fast
'The speed of a plane is faster than that of a train.'

d. Yi ge ren zong yao he shui.
one CL person always will drink water
'A person always drinks water.'

Second, singular predicates reject a numeral but not an article, as seen in the contrasts in (20a) and (20b). When an indefinite article occurs in a predicate, the predicate is a DP (Stowell 1989: 256; 1991). The examples in (21) show that *yi* may occur in such predicates. *Yi* in this use patterns with an article, rather than a numeral.

(20) a. Jill is {a/*one} doctor.
b. As {a/*one} doctor, Jill should know better. (Huddleston and Pullum 2002: 372)

(21) a. Acai shi yi ge xiaoqigui.
Acai be one CL miser
'Acai is a miser.'

b. Zuowei yi ge laoshi, Acai hen naixin.
as one CL teacher Acai very patient
'As a teacher, Acai is very patient.'

Third, the overt numeral *yi* may bear stress, whereas the $\mathrm{YI_D}$ in (19c, d) and (21) never does so. Real numerals are lexical elements but articles are functional ones. The former, but not the latter, easily bears stress. The English article *a/an* does not bear stress in normal conversations (Selkirk 1980 [2015: 34]).

Fourth, the overt numeral *yi* can be associated with a focus marker such as *zhi* 'only', but the $\mathrm{YI_D}$ in (19c, d) and (21) cannot. The contrast is shown in (22). In (22a), *yi* is stressed, and *zhi* must be associated with it, so this *yi* is a numeral. In (22b), however, *yi* is not stressed, and *zhi* is associated with the stressed noun *chushi* 'chef', instead of the closer element *yi*, so this *yi* is not a numeral, but $\mathrm{YI_D}$ instead.

(22) a. Alin zhi gu-le yi wei chushi (bushi liang wei).
Alin only hire-PFV one CL chef not two CL
'Alin hired only one chef (rather than two).'

b. Alin zhi gu-le yi wei chushi. (bushi siji).
Alin only hire-PFV one CL chef not driver
'Alin hired only a chef (rather than a driver).'

Note that plural numerals may also bear stress or be in construal with a focus marker in some contexts and may not do so in some other contexts. But such a flexibility is not found in indefinite articles, which intrinsically reject stress and focus markers.

Fifth, the overt numeral *yi* licenses a numeral-oriented adverb, such as *yigong* 'totally', *zonggong* 'totally', and *zhiduo* 'at most in number', but a non-numeral use of *yi* is not able to do so. In (23a), *yi* licenses *zhiduo*, but the *yi* in (23b) does not. The latter *yi* is $\mathrm{YI_D}$.

(23) a. Alin zhiduo yao gu YI wei chushi, {bushi liang wei
Alin at.most want hire one CL chef not two CL
/#bushi siji}.
/not driver
'Alin wants to hire one chef at most {not two/#not a driver).'

b. Alin (*zhiduo) yao gu yi wei chushi, bushi siji.
Alin at.most want hire one CL chef not driver
'Alin wants to hire a chef (*at most), rather than a driver.'

Indefinite articles in English have different uses (e.g., in predicates and arguments), and so does YI_D in MC. Indefinite articles are banned in certain contexts, and so is YI_D, as already seen in the kind nominals in (2b) and (3b). We thus have evidence for differences between the numeral YI and YI_D in their distributions, their intrinsic phonological and information structure properties, and their ability to license a numeral-oriented adverb. I thus claim that YI_D is a realization of D, in contrast to the numeral *yi*, which is at SpecNuP (see (18)).[10]

Recall that the numeral for 'one' can be either the overt *yi* or the null YI_{pro}. If YI_D occurs with YI_{pro}, the former heads a DP and the latter is at SpecQuantP. Thus, *yi wei chushi* 'a chef' in (22b) has the structure in (24):

(24) $[_{DP[-def]}$ $[_{D'}$ *yi* $[_{NuP}$ YI_{pro} $[_{Nu'}$ ∅ $[_{CLP}$ $[_{CL'}$ *wei* $[_{NP}$ *chushi*]]]]]]]

Let us see how the structure in (24) explains the last contrast presented above: a numeral can license a numeral-oriented adverb, but YI_D cannot. In this structure, YI_{pro} is c-commanded by YI_D. Theoretically, a null numeral should be able to license a numeral-oriented adverb (just as a PRO agent can license an agent-oriented adverb). However, this covert numeral is buried within a DP, as proposed in (24). If a numeral is embedded in a DP, it is unable to establish a dependency with a numeral-oriented adverb. This is similar to the case where a numeral is contained in a nominal that has a demonstrative, *mou* 'certain', or *na* 'which'. In this case, the numeral is also unable to license a numeral-oriented adverb, as seen in (25).

(25) *Alin zhiduo kan-le [na {yi/san} ben shu]?
Alin zhiduo read-PFV which one/three CL book

Indefinites can also be in the order [modifier-*de*-numeral-CL-NP] (see (13b)). A numeral contained in such a form is also unable to license a numeral-oriented

[10] I do not claim that YI_D and *a/an* in English have exactly the same distribution. The latter may occur in a definitional context, as in (i) (Lawler 1973: 109), whereas the former may not, as shown in (ii).

(i) A madrigal is polyphonic.

(ii) (*Yi chu) Gezai-xi shi chuzi Yilan de Taiwan chuantong xiqu.
one CL Gezai-opear be from Yilan DE Taiwan traditional opera
'Gezai-Opera is Taiwanese traditional opera from Yilan.'

adverb, as seen in (26) (cf. Zhang 2015: 420). Thus, only a bare NuP, which does not have any nominal-internal element that is higher than the numeral, can license such an adverb.[11]

(26) *Alin zhiduo kan-le [Amei xihuan de {yi/san} ben shu].
Alin zhiduo read-PFV Amei like DE one/three CL book

We now complete our argumentation for the article or determiner status of YI_{D}. Chierchia (2011: xiv) is right to point out that the existence of CL-languages with determiners is possible. In addition to MC, indefinite articles are also found in other CL-languages. In Lao, for example, a numeral consistently precedes a CL, as seen in (27a) and (27b), but an indefinite article, which has the same form as the numeral for 'one', follows a CL, as seen in (27c). Enfield (2007: 121) reports that "when the numeral *nùng1* 'one' appears after the classifier, it functions as a determiner (i.e., a non-specific marker of a singular entity), appearing in the same slot as demonstratives." The same pattern is seen in Zhuang (Ma 2003: 711), and other Kam-Tai languages (Lu 2012: 142–146).

(27) a. kuu3 sùù4 sòòng3 too3
1SG.B buy two CLF.ANIM
'I bought two (e.g., fish).'

b. kuu3 sùù4 paa3 nùng1 too3
1SG.B buy fish one CLF.ANIM
'I bought one fish.'

c. kuu3 sùù4 paa3 toø nùng1
1SG.B buy fish MC.ANIM one
'I bought a fish.'

Cross-linguistically, it is common for an indefinite singular article to develop from the numeral for 'one' (Givón 1981; Lyons 1999: 105). According to Dryer (2013), the numeral for 'one' is used as an indefinite article in at least 112 languages; Vietnamese is such a language (Simpson et al. 2011: 185). In Turkish, which has no definite article, the numeral for 'one' and the indefinite article share the form *bir*. In German, the article *ein* 'a' looks very similar to the numeral *eins* 'one'. Even in English, the word *one* can be used either as a numeral or as an indefinite determiner (e.g., *She arrived one rainy morning*; see also Kratzer 1998: 177f). Parallel to this, a definite article has developed from a demonstrative in many languages. Jenks (2018: Sec. 6) claims that "Mandarin is at the starting point in this historical chain." The identical forms between the numeral YI and YI_{D} suggest that MC may have similarly developed an indefinite article from the numeral for 'one'.

Like indefinite articles in other languages, YI_{D} has a generalized quantifier interpretation ($<<e, t>, t>$). Indefinites as choice function variables may be existentially closed at any point, and thus they have no constraints on their scope

[11] If *zuiduo* 'at most', instead of *zhiduo* 'at most', occurs in (26), it may be associated with the whole predicate, instead of the numeral. Then the reading will be 'Alin did nothing beyond reading the books.'

interactions with other operators (Reinhart 1997; Winter 1997; Kratzer 1998). The examples in (28) show that YI_D nominals may have either a wide or a narrow scope, a well-recognized property of *yi*-initial indefinites (e.g., Lee 1986; Aoun and Li 1989; Liu 1997).

(28) a. Mei ge ren dou bei yi ge nüren zhuazou le.
every CL man all by one CL woman arrest PFV
'Everyone was arrested by a woman.' $\forall > \exists, \exists > \forall$
(Aoun and Li 1989: 142)

b. Suoyou de laoshi dou jide yi ge xuesheng.
all DE teacher all remember one CL student
'All of the teachers remember one student.' $\forall > \exists, \exists > \forall$
(Liu 1997: 57)

The distinctions between YI_D and the numeral YI suggest that not all nominal-initial *yi*s have the same formal properties or occur in the same structural position. The distinctions also rule out the possibility that an *yi*-initial indefinite is derived by the raising of the numeral YI to D. As we stated in Sect. 2.2, a numeral is base-generated at a Spec position in MC. It is impossible for a Spec element to move to a head position.

In (24), YI_D occurs with YI_{pro}, which is a null proform of the numeral YI. The co-occurrence of an indefinite article with a numeral for 'one' is also seen in languages such as Jingpo. According to Gu (2009), *langai* 'one' can occur in a complex numeral (e.g., *mali-shi-langai* '41'), whereas *mi* 'a' cannot (**mali-shi-mi*); the former, but not the latter, can also occur in an ordinal number, answer a how-many question, and be used in arithmetic. In (29a), in the absence of *mi*, *langai* denotes a pure quantity, but in (29b), in the presence of *mi*, the quantity meaning disappears.[12]

(29) a. Masha langai sha n lo ai.
person one only not enough PRT
'Only one person is not enough.' (We need more.)

b. Masha langai mi sha nga ai.
person one a only have PRT
'There is only a person.' (There is nothing else.) [Jingpo; Gu 2009: 235]

[12] One reviewer asks why the co-occurrence of a numeral and an indefinite article is not seen in languages such as English. I think the semantics of 'one' is available in nominals with *a/an*. According to Krifka (2004: 194; also see Delfitto and Fiorin 2017: 35), unlike other count and mass nouns, singular count nouns are not predicates (properties), but rather are functions from numbers to predicates. They contain a number argument (*n* in (i)), which can be filled by the number word *one*. Thus, we can understand that the indefinite article *a/an* selects a form that has an implicit *one*. But it is not clear to me why this numeral in English and the one occurring with YI_D in MC are silent.

(i) [[dog]] = $\lambda w \lambda n \lambda x$[DOG(w)(n)(x)], = DOG, type $<s, <n, <e,t\gg>$ (Krifka 2004: 192 (52a))

I have argued that MC has an indefinite article, YI_D. But its occurrence is not obligatory for indefinites, since other strategies are available, such as *mou* 'certain' for specific indefinites. Moreover, I have discussed three YIs: the overt numeral YI, which is able to bear stress, a null numeral YI (YI_{pro}), and the unstressed YI, which is an article (YI_D). The first two YIs are at SpecNuP, and the last one heads a DP. Also, the YI_{pro}-containing nominals discussed so far are all DPs, since they have an overt element in DP, such as a demonstrative or YI_D.

3 CL-initial nominals

Having discussed *yi*-initial indefinites, we now turn to CL-initial ones (also called bare CL nominals), which have no overt numerals. I label a CL that occurs at the left-edge of a nominal *iCL* (the letter *i* is for initial). In this section, I discuss four issues: where does the singular meaning of such a nominal come from (Sect. 3.1), where does an iCL surface (Sect. 3.2), how should the analysis developed so far be put in a broader perspective (Sect. 3.3), and how should we understand the syntactic dependency of an iCL nominal on a higher head element in MC (Sect. 3.4)?

3.1 YI_{pro} and the projection of DP

In an iCL nominal, there is no overt numeral, but the nominal consistently has a singular meaning. Since neither the CL nor the NP encodes a singular meaning, there must be a covert element that is responsible for it. I adopt the hypothesis that a null numeral for 'one' occurs in such a nominal (Cheng and Sybesma 1999: 529; Huang 2015: 37; Jiang 2015a: 337, fn. 15). Therefore, the NuP part of the object in (30a) has YI_{pro}, as shown in (30b) (the nominal structure higher than NuP in (30b) is elaborated later in (42b)).

(30) a. Ta na-chu zhang zhaopian.
he take-out CL photo
'He took out a photo.'
b. [… [$_{NuP}$ **YI_{pro}** [$_{Nu'}$ ∅ [$_{CLP}$ [$_{CL'}$ *zhang* [$_{NP}$ *zhaopian*

As I reported in Sect. 2.2, the 'one' reading of YI_{pro} is not focused, and thus it is not allowed to contrast with another numeral. As expected, the 'one' reading of an iCL-nominal is also not focused, and thus it is not allowed to contrast with another numeral either, as seen in (31).

(31) Ta na-chu zhi shoutao, {bushi zhi wazi/ #bushi liang zhi shoutao}.
he take-out CL glove not CL sock not two CL glove
'He took out a glove, {rather than a sock/#rather than two gloves}.'

On the other hand, when the quantity reading is focused, no iCL nominal may occur (Lü 1944 [1990: 167]), as seen in (32) and answer B in (33a). The dialogue in (33a) is similar to the one in (33b). When answering a *how-many* question, a numeral must occur, instead of *a*.

(32) you *(yi) ge xuesheng, wu ge chefu.
have one CL student five CL coachman
'There is one student and five coachmen.' (Lü 1944 [1990: 167])

(33) a. A: Ni you ji ge xiaohai?
you have how.many CL child
'How many children do you have?'
B: Wo you *(yi) ge xiaohai.
I have one CL child
'I have one child.'
b. A: How many children do you have?
B: I have one child./*I have a child. (Krifka 2008: 5)

Moreover, as shown in Sect. 2.2, the 'one' reading of YI_{pro} is never associated with a focus marker; as expected, if the focus marker *zhi* 'only' occurs with an iCL nominal, it is in construal with the NP, instead of the implicit 'one'. This is shown in (34) (also see Jiang 2015a: 337 (46b)). The constructions in (35), which have the focus marker *lian* 'even', demonstrate the same point.

(34) Alin zhi gu-le wei chushi.
Alin only hire-PFV CL chef
A: 'Alin hired only a chef (rather than another kind of person).'
Not B: 'Alin hired only one chef (rather than more than one).'

(35) a. Ta lian *(yi) ge xiaohai dou guanbuliao, geng bie
he even one CL child even not.manage more not
ti liang ge xiaohai le.
say two CL child PRT
'He cannot deal with even one child, let alone two children.'
b. Ta lian ge xiaohai dou guanbuliao, geng bie
he even CL child even not.manage more not
ti guan daren le.
say manage adult PRT
'He cannot deal even with a child, let alone adults.'

An iCL nominal thus denotes a 'one' reading, but this quantity must not be focused. Zhang (2013: 142–143) further shows that such a nominal may not be in construal with *zuzu* 'as many as', *zhengzheng* 'in total', and cannot be selected by a V-*man* 'V-full' verb. All of these indicate that it is YI_{pro} that is responsible for the non-focused 'one' meaning of iCL nominals.

If iCL-nominals had a reduced nominal structure, such as a bare CLP (cf. Cheng and Sybesma 1999: 528 for Cantonese), their consistent singular reading would come from nowhere.[13] Also, since YI_{pro} occurs with iCL, the whole nominal cannot be a bare NuP (called NumberP in some literature), either. This is because the latter can be focused and thus encode an "exactly" reading of the numeral (cf. Li 1998, Tsai, to appear), whereas an iCL nominal, like an YI_{D} nominal, cannot do so. This means that the nominal has to be a projection higher than NuP, i.e., it is DP. In other words, the occurrence of YI_{pro} implies the projection of DP. In MC, iCL nominals contain YI_{pro}, and thus they are DPs.

In Sect. 2.2, we saw that YI_{pro} occurs in the presence of a D-domain element, such as a demonstrative. In Sect. 2.4, we saw that YI_{pro} occurs with YI_{D}. Now we see that YI_{pro} also occurs in a DP that has an iCL. In all three cases, YI_{pro} occurs in a DP. But an iCL is not a base-generated D element or article in the DP. First, every CL can be used as an iCL, and there are dozens of CLs in a language; it is unlikely for a language to have dozens of articles. Second, articles do not s-select nouns, but many CLs do (Zhang 2013: 163). Third, in languages such as Cantonese and Wu, an iCL nominal can be either definite or indefinite (e.g., Cheng and Sybesma 1999; Li and Bisang 2012: 336). It is unlikely for the same form to be both a definite and an indefinite article. As pointed out by Simpson (2005: 824), CLs do not encode definiteness intrinsically, while base-generated D elements do.

If an iCL is hosted by a DP and the iCL itself is not base-generated at the D head, the DP has neither a base-generated head element nor an element at the Spec position. Theoretically, the Spec and the head of a syntactic projection cannot both be null: "A projection is interpretable iff it is activated by lexical material" (Koopman 1996: (56)). Accordingly, I claim that iCL has moved to D, activating a DP. This theoretical argument is backed up with empirical arguments, to be presented in Sect. 3.2.[14]

3.2 The surface position of the initial CL

In this subsection, I present evidence for the raising of iCLs to D. An iCL shows properties of a D-element, rather than properties of a CL that follows an overt numeral, and the D-properties indicate that an iCL has reached a D position. The first parallelism between D-head elements and iCLs is seen in NP-ellipsis. Recall

[13] Arguing that iCL indefinites are semantically different from *yi*-initial ones in MC, Li and Bisang (2012: 357) claim that the former are bare CLPs. Jiang (2012) falsifies their argumentation. Also see (43) and (44) below.

[14] The plural indefinite subject in (i) has no overt element in DP. Note that the null D in (i) does not need any overt head element licensor (see Simpson 2005: 830 for relevant discussion; cf. Contreras 1986; Longobardi 1994). In this respect, MC is not different from other languages (see the English translation of (i)). Theories on how DP is lexicalized in plural nominals in English may apply to MC (cf. Koopman 1996: sec. 3.6 for an analysis of other functional projections that do not seem to have overt material).

(i) Liang ge pangpang de xiaohai zou-le-guolai.
two CL fat DE child walk-PFV-over
'Two fat kids came over.'

that numerals in MC, as phrasal elements, do not license ellipsis ((14a)). The article *a* in English does not license ellipsis of the element to its right (**I wanted to read a book, so I bought a*; Elbourne 2005: 46). $\mathrm{YI_D}$ does not do so, either, as seen in (36). It has been recognized that not all functional heads license ellipsis. For example, the infinitive *to* licenses VP ellipsis in a control construction but not in an ECM construction (e.g., Lasnik 1992). Thus, an article's inability to license ellipsis is not surprising.

(36) Alin gu-le yi wei chushi, bushi siji. *Tamen ye gu-le yi.
Alin hire-PFV one CL chef not driver they also hire-PRE one
Intended: 'Alin hired a chef, rather than a driver, and so did they.'

Now consider the minimal pair in (37). In (37a), in the presence of *yi*, the CL *ben* does not move and is able to license ellipsis. In contrast, in (37b), in the absence of *yi*, the same CL loses its ability to license ellipsis (Lü 1990 [1944]: 170. Sec. (b)). Thus, an iCL and an article are similar in this respect. It is possible that when a CL adjoins to D, since D does not license ellipsis, the whole CL-D cluster becomes unable to license ellipsis.[15]

(37) a. Ta mai-le yi ben shu, wo ye mai-le yi ben.
he buy-PFV one CL book I also buy-PFV one CL
'He bought a book, and so did I.'

b. Ta mai-le ben shu, *wo ye mai-le ben. (Yang 2001: 76)
he buy-PFV CL book I also buy-PFV CL

The second parallelism between D-head elements and iCLs is seen in head-stranding (by either a movement trace or empty element). Like the article *a/an* in English, $\mathrm{YI_D}$ cannot be stranded, as seen in (38).

(38) Chushi, women bu-que. Siji, women dao xiang gu yi *(wei).
chef we not-lack driver we rather want hire one CL
'Chefs, we do not lack. As for drivers, we'd like to hire one.'

Now consider the minimal pairs in (39). In (39a), in the presence of *yi*, the CL *ben* can be stranded. In contrast, in (39b), in the absence of *yi*, the same CL cannot be stranded. Thus, an iCL and an article are similar in this respect. It is possible that when a CL adjoins to the null D, since D head elements cannot be stranded, the whole cluster CL-D cannot be stranded, either.

[15] Note that in an iCL nominal, the ellipsis of the post-*de* part is licensed by *de*, as seen in (i) (also see (13a)).

(i) Mai ge hao de ba!
buy CL good DE IMP
'Buy a good one!'

(39) a. Shu, ta mai-le yi ben.
book he buy-PFV one CL
'As for books, he bought one.'
b. *Shu, ta mai-le ben. (Yang 2001: 75)
book he buy-PFV CL

The third parallelism between D-head elements and iCLs is seen in bearing a contrastive focus. As shown in Sect. 2.4, like the article *a/an* in English, YI_D never bears a focus. Now consider the minimal pair in (40). In (40a), in the presence of *yi*, the CL *ba* in the second clause can contrast with the CL *chuan* in the first clause (the *yi* here is an YI_D).[16] However, in (40b), in the absence of *yi*, the same CL cannot bear a focus. Thus, an iCL and an article are similar in this respect. It is possible that when a CL adjoins to D, since D is unable to bear a contrastive focus, the whole cluster CL-D is not able to do so, either. The examples in (41) show that in the absence of a contrastive focus, the iCL-nominal *ba yaoshi* is well-formed.[17]

(40) a. Ta na-chu yi chuan yaoshi, bu shi yi ba yaoshi.
he take-out one bunch key not be one CL key
'He took out a bunch of keys, rather than one key.'
b. Ta na-chu chuan yaoshi, *bu shi ba yaoshi.
he take-out bunch key not be CL key
Intended: 'He took out a bunch of keys, rather than one key.'

(41) a. Ta na-chu ba yaoshi.
he take-out CL key
'He took out a key.'
b. Na bu shi ba yaoshi ma?
that not be CL key Q
'Isn't that a key?'

In all of the above constructions in this subsection, changing the post-CL noun from monosyllabic to disyllabic or vice versa does not affect the acceptability pattern, and thus the issue is not prosodic. The above three points show that iCLs are similar to D head elements and different from non-initial CLs. They suggest that

[16] Yang (2001: 72) states that phonologically a CL is never stressed. But the CL in (40a) is stressed. Such an example shows that the statement is not accurate.

[17] In (i), which is from a reviewer, the nouns *ren* 'person' and *baishe* 'decoration' each bear a contrastive stress, but the CLs *ge* and *jian* do not.

(i) Wo shi ge ren, bu shi jian baishe.
I be CL person not be CL decoration
'I am a person, not a decoration.'

iCLs do not remain in CLP. I claim that an iCL has moved from CLP to DP. Accordingly, the structure of the CL-initial object in (42a) (= (30a)) is (42b).

(42) a. Ta na-chu zhang zhaopian.
he take-out CL photo
'He took out a photo.'
b. $[_{DP}\ [_{D'}\mathit{zhang}\ [_{NuP}\mathrm{YI}_{pro}\ [_{Nu'} < \mathit{zhang} > [_{CLP}\ [_{CL'} < \mathit{zhang} >$ $[_{NP}\ \mathit{zhaopian}]]]]]]]$

My arguments for the iCL-raising rule out not only a bare CLP analysis, but also an YI$_{pro}$-raising analysis of iCL nominals. One might assume that it is YI$_{pro}$ that moves to DP, leaving the CL in situ. We have seen that a CL behaves differently in the a-forms and the b-forms in (37), (39), and (40). The YI$_{pro}$-raising analysis is unable to explain these differences. YI$_{pro}$ is structurally higher than a CL, and thus the raising of the former should not affect the properties of the latter.

Since an iCL moves to D, it surfaces at the same position as YI$_{D}$. YI$_{D}$ nominals and iCL nominals are both hosted in DP. They are also semantically similar (Li and Feng 2015: 2 fn.1). For example, both exhibit scope ambiguity, as seen in (43a) (Jiang 2015a). I also notice that both may have a generic reading (contra Simpson and Ngo 2018: 241), as seen in (43b), and both reject a kind-reading, as seen in (43c).[18] Also, both may occur in positions that exclude nonspecific nominals, as seen in (44) (= (5)). I thus claim that YI$_{D}$ and iCL-raising are two ways to realize the indefinite D.

(43) a. [ruguo ni neng dai (yi) ge nüsheng lai wode
if you can bring one CL girl come my
paidui dehua], ni he duoshao wo dou mai-dan.
party if you drink much I all pay-bill
'If you can bring a girl to my party, no matter how much you drink I will pay for it.' *a girl* > *if, if* > *a girl* (Jiang 2015a: 336)
b. Na zhong ma pao de bi (yi) liang che hai kuai.
that kind horse run DE than one CL car even fast
'That kind of horse runs faster than a car.'
c. *Renmen ba (yi) zhi jingyu gui-wei buru-dongwu.
people BA one CL whale classify-as mammal
Intended: 'People classify whales as mammals.'

(44) a. Ta ba (yi) ge hao pengyou gei dezui le.
he BA one CL good friend give offend PFV
'He got a good friend offended.' (cf. Huang 2015: 35)
b. Ta jiao-le (yi) ge nü-pengyou hen piaoliang.
he make-PFV one CL girl-friend very pretty
'He has a girlfriend who is very pretty.' (Jiang 2015a: 336)

[18] Bare CL nominals also allow a generic reading and reject a kind reading in Yi (Jiang 2015b: 56).

In Sect. 2.2, I gave a few arguments against the *yi*-deletion analysis on numeralless CL-nominals. An *yi*-deletion analysis of iCL nominals has been assumed by Lü (1990 [1944]: 166), Chao (1968: 554), Wang (1989: 109), among many others. One problem with this analysis is that the part assumed to be elided has no antecedent (Zhang 2013: 140–143). In (45), if *wu* 'five' occurs in the first clause and qualifies to be an antecedent for an assumed deletion, the deleted numeral in the second clause should be *wu*. However, the implied numeral meaning in the second clause is consistently 'one'. My other arguments against *yi*-deletion in 2.2 also apply to iCL nominals.

(45) Baoyu na-chu wu ge pingguo, Daiyu ye na-chu ge pingguo.
Baoyu take-out five CL apple Daiyu also take-out CL apple
'Baoyu took out five apples, and Daiyu also took out an apple.'

I conclude that iCL nominals are derived by CL-to-D movement, with $\mathrm{YI}_{\mathrm{pro}}$ in SpecNuP. Such nominals are not derived by *yi*-deletion and they are not bare CLPs, either.

3.3 The null numeral for 'one' and bare CL nominals again

The projection of DP for iCL nominals depends on the availability of $\mathrm{YI}_{\mathrm{pro}}$ (Sect. 3.1). Cross-linguistically, a bare CL nominal is consistently singular, indicating the occurrence of a null numeral for 'one', which implies the occurrence of a projection that is higher than NuP, i.e., DP. The raising of an iCL serves to activate the projection of DP that contains $\mathrm{YI}_{\mathrm{pro}}$, in order to satisfy Full Interpretation (see Sect. 3.1). My analysis predicts that bare CL nominals occur if and only if a null numeral for 'one' occurs in the language. This is indeed the case. All of the languages introduced in Sect. 2.2 that allow the direct combination of a D-domain element with a CL, without an overt numeral, allow bare CL nominals: Cantonese and Vietnamese (e.g., Cheng and Sybesma 1999: 524; Simpson et al. 2011), Thai, Zhuang (Lu 2012: 67–68), Lao (Enfield 2007: 131, with *qiik5* 'more'), Nuosu Yi (Jiang 2015a: 328), and Bangla (Dayal 2012). In contrast, all of the languages that disallow such a direct combination also disallow bare CL nominals: Korean and Japanese (Satoshi Tomioka and Byeong Yi p.c.; Jiang 2015a: 325), and Min (e.g., Chen 1958: 503; Zhou 1991: 212; Cheng and Sybesma 2005: 268 (20); Li and Feng 2015).[19] However, we have not seen a language that has bare CL-nominals but rejects the direct combination of a demonstrative with a CL (i.e., it does not have a null form for 'one'). Cheng and Sybesma (2005: 287) attribute the lack of a bare CL nominal in Min to the fact that the numerals in the language cannot be empty. I propose a similar generalization in (46).

(46) If a language allows bare CL nominals, it also allows the direct combination of a CL with an overt D-domain element such as a demonstrative, without an overt numeral.

[19] According to Cheng & Sybesma (2005: 290 fn. 6) and my field work, young speakers of Taiwan Southern Min accept some bare CL nominals, under the influence of MC.

Huang (2015: 37) also links the syntax of bare CL nominals to the syntax of the numeral for 'one'. He assumes that the numeral for 'one' occurs in the Num head, and further assumes that first, Num is strong in Cantonese, and thus a CL moves to Num and then to D, deriving a bare CL nominal; second, since Num is not strong in MC, iCL remains in situ; and third, since Num is lexical in Taiwan Southern Min, no CL moves to Num. He further assumes that the head movement correlates with the definite reading of the nominal, and thus the movement is optional in Cantonese. When the head movement does not occur, the bare CL nominal is indefinite, as seen in MC and the indefinite bare CL nominals in Cantonese (see his footnote 36). However, we see no evidence for the head status of numerals in these languages (see (14)), and the non-D surface position of iCLs in MC does explain the facts reported in Sect. 3.2 here. Thus I do not adopt this version of a head movement analysis for bare CL nominals. But Huang (2015), Cheng and Sybesma (2005), and the present study all see the relation between the existence of bare CL nominals and the availability of a null form of the numeral for 'one'. I consider this availability a necessary condition for the CL-to-D movement.

CLs are nominal auxiliaries (called "auxiliary nouns" in Chao 1948, 1968: 584 fn. 26; Zhang 2013: 208–211, and light nouns in Huang 2015: 8). In the verbal domain, the alternation between a base-generated functional head element and a raised auxiliary is well recognized. For instance, in English, a conditional clause can be introduced either by the complementizer *if* or by the raising of an auxiliary. The two examples in (47) mean the same thing. It is not surprising that nominal auxiliaries are also able to undergo head movement to D, alternating with an article.

(47) a. If I had any inkling of this, I would have acted differently.
b. Had I any inkling of this, I would have acted differently. (Huddleston and Pullum 2002: 753)

Bare CL nominals are indefinite in MC and Yi (Jiang 2015a: 328), definite in Bangla (Dayal 2012), and either definite or indefinite in Cantonese, Wu, and Vietnamese (e.g., Cheng and Sybesma 1999: 524; Li and Bisang 2012). They all denote a singular meaning, indicating the existence of a null numeral for 'one', which in turn implies the projection of DP and a CL-to-D raising. I have argued for CL-to-D raising for indefinite bare CL nominals in MC. For definite bare CL nominals in other languages, Simpson (2005), Li and Bisang (2012), and Huang (2015) also argue for CL-to-D raising. It is possible that the definiteness of such nominals comes from factors additional to the necessary condition for the CL-raising mentioned above. From the perspective of the building of a complex expression, it is common for a certain type of head movement to be subject to different conditions cross-linguistically. From the perspective of definiteness, it is common for the same form to be either definite or indefinite. For example, bare nouns are either definite or indefinite in MC and Tagalog (Collins 2018), but not definite in languages such as English.

Moreover, the availability of CL-to-D raising means that DP is projected in CL languages. If bare CL nominals are rejected in a language, bare CLPs cannot be arguments. On the other hand, if such nominals are allowed in a language, they are

DPs, instead of bare CLPs. Either way, bare CLPs cannot be arguments in CL languages. Therefore, a CLP is always contained in a higher projection and thus it does not have an independent syntactic status (contra Cheng and Sybesma 1999).

3.4 The dependency of an initial CL on a higher head element in MC

Bare CL nominals are subject to a syntactic condition that is not seen in other nominals in MC. This condition has been claimed as a government relation (e.g., Li and Feng 2015: 4, 9 fn. 10; Huang 2015: 35). The basic idea is that an iCL nominal must be c-commanded by an overt head element without any intervening clausal or nominal node (see Li and Feng 2015: 4). The iCL nominals in (48) all satisfy this condition. The head element is the verb *song* 'give' in (48a), the preposition *cong* 'from' in (48b), the head of FocusP *lian* 'even' in (48c), and *ba* in (48d).

(48) a. Wo hui song tamen <u>ge liwu</u>. (Li and Feng 2015: 4)
I will give they CL gift
'I will give them a gift.'

b. Ta cong <u>ge hezi-li</u> na-chu kuai tang lai. (ibid. p. 3)
he from CL box-in take-out CL candy come
'He took a candy out of a box.'

c. Ta lian <u>zhang zhi</u> dou bu gei xuesheng.(ibid. p. 7)
he even CL paper even not give student
'He did not give the students even a piece of paper.'

d. Ta ba <u>ge hao pengyou</u> gei dezui le.
he BA CL good friend give offend PFV
'He got a good friend offended.' (Huang 2015: 35)

In (49a), however, the iCL nominal *zhi gou* is not governed by any head element. In (49b), the iCL nominal *ge hao zuojia* is contained within a nominal. Its closest overt c-commanding head element is *zhao* 'seek', but its relation to *zhao* is separated by a nominal node (marked as DP by me). In (49c), the iCL nominal *ge xuesheng* is contained in a clause. Its closest overt c-commanding head element is *ruguo* 'if', but its relation to *ruguo* is separated by a clause node (marked as IP by me). Similarly, in (49d), the iCL nominal *ge ren* is contained in a clause. Its closest overt c-commanding head element is *zhiwang* 'expect', but its relation to *zhiwang* is separated by a clause node. In all of these cases, the iCL nominals are not acceptable.

(49) a. Waimian *(yi) zhi gou zai jiao.
outside one CL dog PRG barking
'Outside, a dog is barking.' (adapted from Yang 2001: 32 (31))
b. Ta xiang zhao [$_{DP}$ [*(yi) ge hao zuojia] de shu].
he want seek one CL good writer DE book
'He is looking for books by a good writer.' (Li and Feng 2015: 4)
c. ruguo [$_{IP}$ [*(yi)ge xuesheng] bing-le]
if one CL student ill-PFV
'if a student gets ill'
d. Wo zhiwang [$_{IP}$ [*(yi)ge ren] bangzhu wo].
I expect one CL person help me
'I expect a person to help me.'

Although this version of government is able to cover examples like those in (48) and (49), it fails to cover examples like those in (50) and (51a), where the iCL nominal is governed by an overt head element in each case, but it is still unacceptable. The head element is the coordinator *he* in (50a) (Li and Feng 2015: 4; Yang 2001: 69), and the post-modifier *de* in (50b) and (50c) (see Zhang 2015 for the head status of *de* in these constructions). The same head form *dui* 'to' can be followed by the iCL nominal *ge xuesheng* 'a student' in (51b), but not in (51a).

(50) a. Ta xiang mai ben shu he *(yi) zhang zhi.
he want buy CL book and one CL paper
'He wants to buy a book and a piece of paper.' (Lü 1990 [1944]: 167)
b. houhou de *(yi) ben shu
thick DE one CL book
'a thick book' (adapted from Yang 2001: 69)
c. Wo zai zhao [xue yingyu de *(yi) ge xuesheng].
I PRG seek study English DE one CL student (= (13b))
'I am looking for a student who studies English.' (adapted from Li and Bisang 2012: 346)

(51) a. Ta dui *(yi) ge xuesheng hen shiwang.
he to one CL student very disappointed
'He is disappointed in a student.'
b. Ta dui (yi) ge xuesheng ju-le yigong.
he to one CL student bow-PFV once
'He bowed towards a student.'

Li and Feng (2015) explore the prosodic and stylistic constraints on iCL nominals when the syntactic constraint is satisfied. Complementing their research, one can try to understand the syntactic part: how to distinguish good head elements, such as those in (48), from bad ones, such as those in (49b-d) and (50), and why a head element must occur, as shown by (49a).

Regarding the first issue, one hypothesis compatible with Baltin's (2012: 400) incorporation view on government is that an iCL nominal and a higher head element establish a semantic relation, forming a semantically enriched complex, which must be interpretable and compatible with the intended meaning of the speaker. The complex established in this way is the meaningful *song ge liwu* 'send a gift' in (48a), the source-denoting *cong ge bezi-li* 'from a box' in (48b), the focused nominal *lian ge xuesheng* 'even a student' in (48c), and the causee topic *ba ge hao pengyou* 'a good friend' in (48d). But for a coordinator-containing expression to be interpretable, it must have at least two conjuncts. Thus, the combination of the coordinator *he* and the second conjunct *zhang zhi* in (50a) does not lead to an interpretable complex.[20] In (50b), the complex *de ben shu* is also not interpretable. (50c) can be explained in the same way. In (51a), *dui ge xuensheng* 'a student' is not semantically richer than the form without *dui* in this syntactic context, since the occurrence of *dui* is purely syntactic (Huang et al. 2009: 22). However, in (51b), *dui ge xuensheng* 'towards a student' encodes a goal, which is indeed semantically richer than the form without *dui*. In (49b), although the combination of *zhao* 'seek' and *ge hao zuojia* 'a good writer' is meaningful, this meaning is not what is intended. In (49c), the combination of *ruguo* 'if' and *ge xuesheng* 'a student' is not interpretable. In (49d), although the combination of *zhiwang* 'expect' and *ge ren* 'a person' might be meaningful, this meaning is not what is intended. Thus, the semantic grouping is able to explain why an iCL nominal is disallowed in these various constructions.

This grouping of an iCL nominal with a higher head element can be understood as a semantic reanalysis. This reanalysis is parallel to morphological or prosodic reanalyses, which establish morphological or prosodic clusters. In all of these different types of reanalysis, constituency may be changed to satisfy a certain consideration. For example, the constituency of the BA construction in (48d) is (52a) (see Li 2006: 412), but the semantic reanalysis groups *ba* and the iCL nominal into a semantic unit, at LF (the shaded string). Also, in (50a), the first conjunct *ben shu* 'a book' is an iCL nominal, which does not form a constituent with *mai* 'buy', as shown in (52b), but the semantic reanalysis groups them together at LF.

(52) a. [$_{\text{baP}}$ <ta> [$_{\text{ba'}}$ ba [$_{\text{vP}}$ ge hao pengyou [$_{\text{v'}}$ gei dezui-le [$_{\text{VP}}$ …]]]]] (48d)
he BA CL good friend give offend-PFV

b. [$_{\text{VP}}$ mai [$_{\text{\&P}}$ [ben shu] he [yi zhang zhi]]] (50a)
buy CL book and one CL paper

[20] In (i), the iCL nominal *ge baozi* 'CL bun' follows the conjunction *haiyou* 'and'. The example is judged acceptable in Jiang (2012: 212), but it is not acceptable to me and my informants.

(i) Gei wo lai ge chaye-dan haiyou ge baozi.
Give me come CL tea-egg and CL bun
'Give me a tea egg and a bun.'

The second issue is why this semantic reanalysis with a head element is required, as seen in (49a). Since an iCL nominal requires the occurrence of another element, the dependency can be understood as a certain type of selection, and since the dependency is at LF, the selection does not have to be between sisters in their surface positions (cf. Bhatt and Pancheva 2004). The selection reminds us of Bruening's (2010: 534) hypothesis that a phrase also selects another element (e.g., an AP c-selects a nominal and an AdvP c-selects a verbal expression), but the selector does not project in this non-canonical selection. Theoretically, then, it is possible for an iCL nominal to select another element, and if the selection is not satisfied in a given construction, the DP is not acceptable in it. However, this non-canonical selection is also non-canonical in two further senses. First, the selection is neither c-selection nor s-selection. Instead, it is category-level selection: an iCL nominal selects a head element (of any category). The distinctions between different category levels are significant syntactically. Elements at the minimal level (i.e., head elements) intrinsically decide the category label of the merge, whereas those at other levels do not (Chomsky 2013: 43). Second, the selection is backward: an iCL nominal, as a selector, is c-commanded by the selected head element. Backward syntactic relations in various senses have been discussed in the literature, for example backward binding (e.g., Belletti and Rizzi 1988; Platzack 2012), backward control (Polinsky and Potsdam 2002; Potsdam 2009), and backward raising (Potsdam and Polinsky 2012). The existence of backward selection seems plausible, although it has not been explored in the literature yet. I leave study on this possibility to future work.

The semantic reanalysis hypothesis explains cases that the government theory does not explain. Like other non-canonical dependencies discovered in syntax, the one exhibited in bare CL nominals in MC is not universal. Bare CL nominals in Yi must also be indefinite, but they can occur as subjects and thus do not exhibit this dependency (Jiang 2015b: 58).

In this section, I have argued for the existence of $\mathrm{YI}_{\mathrm{pro}}$ in an iCL nominal, argued for a CL-to-D movement in deriving such a nominal, linked the availability of such a nominal to the availability of a null numeral for 'one' in a language, compared the movement of a CL in the nominal domain to the auxiliary movement in the verbal domain, and proposed a new semantic reanalysis hypothesis to explain the dependency of such a nominal on a head element in MC.

4 Conclusions

I have reached the following major conclusions in this paper. First, MC has a null proform for the numeral for 'one'. Second, *yi* can be used as an indefinite article in MC, and thus DP exists in the language. Third, a bare CL nominal is derived by a CL-to-D movement, and thus such a nominal is a DP, rather than a bare CLP. Fourth, a semantic LF reanalysis is able to explain the dependency of an iCL nominal on a higher head in the language.

CL languages may have definite articles, as in Indonesian (Winarto 2016; Little and Winarto 2018) and Nuosu Yi (Jiang 2018), and indefinite articles, as in Lao (Enfield 2007) and Jingpo (Gu 2009). This paper shows that MC may also have indefinite articles. Moreover, Turkish has an indefinite article but no definite article, whereas Hebrew has a definite article but no indefinite article (Doron 2003). Cross-linguistically, whether a language has a particular type of article does not correlate with whether the language is a numeral CL language or not. DP headed by an article is found in various types of languages. Thus there is no reason to assume that MC is an articleless language.

This study also shows that the head of a DP can also be realized by a raised CL in a CL-language. Thus, the projection of DP is attested in various ways in various types of languages. It is not true that a CL-language cannot have DP. The research supports Jenks's (2018: Sec. 6) claim that complex nominals involve full DP structure. "As such, the debate about whether a language is a 'DP language' or an 'NP language' is not on quite the right track. Instead, we must ask under which semantic contexts languages are required to project DP."

Furthermore, since bare CL nominals are argued to be DPs, a bare CLP is not able to function as an argument in a language, unlike the projection that hosts a numeral (NuP) and DP. It is possible that a bare CLP is not a phase category.

Finally, explaining the dependency of an initial CL on a higher head element in MC may lead us to uncover more non-canonical dependencies in syntax.

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